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Rahul Patwadi

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San Jose, CA, US

Education

San José State University, San Jose, CA *Aug 2026 – May 2028*
M.S., General Engineering (Software / AI Systems Engineering)

JSS Academy of Technical Education (VTU), Bangalore, India *2018 – 2022*
B.E., Electronics and Communication Engineering

Work Experience

Senior Analyst (Software / GenAI Engineer), Dell Technologies *Feb 2026 – Aug 2026*
Bangalore, India

- Independently designed and built an end-to-end **Python/FastAPI** monitoring service for Dell's production Explainability SRE platform, tracking 8 daily source files across two batch windows and alerting engineers in real time to help prevent downstream SLA breaches.
- Designed a two-table **PostgreSQL** schema with state tracking for exactly-once email alerts, replacing manual chat-based follow-ups and reducing file-delay detection from 1–2+ hours to near real time.

Analyst (Software / GenAI Engineer), Dell Technologies *Jun 2025 – Jan 2026*
Bangalore, India

- Contributed to a **five-agent agentic AI workflow** using **LangChain ReAct agents** for root-cause analysis and remediation across 37 production Control-M jobs, spanning observability, analysis, root cause, solution recommendation, and auto-fix agents.
- Designed a **human-in-the-loop approval gate** before automated fixes and expanded incident tracking from two to three states, strengthening workflow governance and improving the accuracy of production SRE reporting.
- Built a full-stack **Python/Angular** monitoring application integrating live Control-M status for 37 jobs across 2 AM/5 PM cycles and triggering the agentic workflow on failures, replacing manual Excel tracking and saving an estimated 40+ SRE hours per month.

Analyst (.NET / Software Engineer), Dell Technologies *Jun 2023 – May 2025*
Bangalore, India

- Implemented synchronous/asynchronous document-processing flows in AV Scanner, a **.NET Web API**; contributed to a Windows-VM-to-**Kubernetes** migration and testing that raised code quality from 58% to 90%.
- Built **FastAPI REST APIs** and an Angular frontend for the Purchase Order Assistant, an ML-based PDF data-extraction tool, and maintained production microservices using **CI/CD pipelines**, Docker, SonarQube, and Tenable.

Projects

NASA Mission Intelligence
Python, FastAPI, Angular, ChromaDB, Gemini API, Docker, Google Cloud Platform

- Built an end-to-end **retrieval-augmented generation (RAG)** system across six NASA missions, ingesting 539 of 545 records into 86,003 indexed chunks through an asynchronous document harvester.
- Built a retrieval evaluation harness with a hand-labeled relevance set; an embedding-dimensionality trade-off improved mean precision@5 from 0.67 to 0.70 while reducing vector storage from 1.5 GB to 742 MB.
- Containerized the full-stack application with **Docker** and deployed it on **Google Cloud Platform**. → **Live Demo**

Autonomous Network Operations Incident Coordination
Python, FastAPI, Pydantic, SQLite

- Built the decision-making core of a **multi-agent incident-response system** using an Expected Value model to triage signals (suppress, verify, escalate, broadcast).
- Built a risk model penalizing suppression of real incidents; achieved 85% decision accuracy on a hand-labeled evaluation set.

Technical Skills

- **Languages:** Python, C#, TypeScript, SQL
- **AI/ML:** Agentic AI, LangChain, **LangGraph**, Multi-Agent Orchestration, ReAct Agents, RAG, Prompt Engineering, Gemini API
- **Backend & Frontend:** FastAPI, .NET (ASP.NET), Angular, REST APIs, Microservices, Pydantic
- **Data & Cloud:** PostgreSQL, SQLite, ChromaDB, Google Cloud Platform, Docker, Kubernetes, Helm
- **DevOps & Tools:** GitLab CI/CD, Git, Linux, SonarQube, Jira, Confluence
- **AI Development Tools: Model Context Protocol (MCP)**, **Claude Code**, **GitHub Copilot**